

P017

Title: Optimizing MEC Migrations for Energy and User Quality of Experience 1

Supervisor: Liam Murphy

Tags: Mobile/wireless networking, performance evaluation, simulation, iFogSim2

Theme: Software Engineering and Distributed Systems

Stream: Both

Description: Cloud computing has undergone phenomenal growth in recent years, but one of its major disadvantages is the distance between cloud servers and end-users. This problem has led to the development of Multi-Access Edge Computing (MEC) which aims to place cloud resources closer to the end-user. Previous projects explored the trade-off between energy efficiency and a user's Quality of Experience (QoE) by examining different initial placement strategies for application modules in the fog computing simulation toolkit, iFogSim2. This project will continue to develop this approach e.g. by considering other models of QoE and/or energy consumption, and/or other sources of variation in the system.

Mandatory deliverables:

1. Conduct a comprehensive literature review of QoE and Energy Efficiency in MEC.
2. Propose, refine and specify a simulation or prototyping solution to an issue in this area.
3. Basic implementation and evaluation of the proposed solution in 2.

Discretionary deliverables:

1. More realistic implementation of the proposed solution.
2. Comprehensive evaluation of the proposed solution.

References:

1. Satyanarayanan, M. The Emergence of Edge Computing. Computer (Volume: 50, Issue: 1, January 2017).
2. Mach, P. & Becvar, Z. Mobile edge computing: A survey on architecture and computation offloading. IEEE Communications Surveys & Tutorials (Volume: 19, Issue: 3, 2017).

3. ETSI. Multi-access Edge Computing (MEC): Framework and Reference Architecture. ETSI GS MEC 003 V4.1.1 (2025-05).
4. Taleb, T. et al. On Multi-Access Edge Computing: A Survey of the Emerging 5G Network Edge Cloud Architecture and Orchestration. IEEE Communications Surveys & Tutorials (Volume: 19, Issue: 3, 2017).
5. Abbas, N., Zhang, Y., Taherkordi, A. & Skeie, T. Mobile Edge Computing: A Survey. IEEE Internet of Things Journal (Volume: 5, Issue: 1, February 2018).
6. Mao, Y., You, C., Zhang, J., Huang, K. & Letaief, K. B. A Survey on Mobile Edge Computing: The Communication Perspective. IEEE Communications Surveys & Tutorials (Volume: 19, Issue: 4, 2017).
7. Mahmud, R. et al. iFogSim2: An extended iFogSim simulator for mobility, clustering, and microservice management in edge and fog computing environments. Journal of Systems and Software, Volume 190, August 2022.